

PANKAJ ARVIND DESHPANDE

pankaj_a_deshpande@yahoo.co.in, +91-7083631680

302, A-7, Kumar Prerana, Near Medipoint Hospital, Aundh, Pune – 411007, Maharashtra, India.

<http://www.linkedin.com/in/pankajAdeshpande>

SUMMARY

A communication systems engineer with 18 years of experience as individual software developer for terrestrial cellular technologies like 3GPP (2G, 3G, 4G), WiMAX and wire-line technologies like ADSL2/2+, and with a solid foundation in Mathematics, Signal Processing with desire and ability to learn.

SKILLS

Skill Type	Skill Particulars
Knowledge Areas	3GPP Access technologies – especially about Layer 1 in 5G (NR), 4G (LTE), 3G (UMTS), 2G (GSM/EDGE)
System Performance Tools	e-BPF, perf-events, Intel oneAPI, Intel vTune
Programming languages	C on Linux with ICC, GCC
	C++ (upto C++17)
	Shell for Bash
	Python
Signal Processing Tools	Matlab, Octave, R
	Assembly language for VLIW ISA
Processors/Architectures	x86 Xeon, DSPs from Texas Instruments, Ceva
Work environments	Linux – Cent OS 7, Ubuntu 20
Containers	LxC, Dockers, Kubernetes – Beginner as a user
Version Control Tools	GIT, IBM Clearcase, SVN, CVS
Programmer's Editors	Vim 8 with cscope integrated
Computer network tools	Wireshark Protocol analyser
Instruments/Masurement tools	Agilent and Rhode & Schwartz Spectrum Analyzers, Signal generators, Oscilloscope, Spirent ADSL Loop Simulators and Noise Generators, Wireless Channel Simulator (SOFI), GSM Signal Generator (SMIQ)
Miscellaneous tools	gdb, Beyond Compare, Latex, Doxygen, MS Excel, Message Sequence Chart Generator

WORK EXPERIENCE

1) Mavenir Systems Pvt. Ltd., Pune; (1y5m)

Associate Principal Engineer, Systems Engineering; (October 2022 – Present)

- Interaction with PLM team to know and understand the customer requirements
- Study and analysis of the requested feature and creation of requirements document and suggestion of testing for the feature
- Interaction across functional teams to collect information pertaining to new feature requests from customers

Tasks	• Dimensioning of mmWave support feature – estimation of number of cores needed to support mmWave on existing FlexRAN L1 code • Analysis of memory optimization delivered in FlexRAN v22.03 • Refactoring of Matlab simulations for NB-IoT programming • Collection of knowledge in the company for supporting customer feature request e.g. improvement in receiver sensitivity, etc. • Exploring system performance evaluation tools
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**2) Parallel Wireless India Pvt. Ltd. (<http://www.parallelwireless.com>), Pune; (8y6m)
Sr. Tech. Lead, (January 2014 – October 2022)**

	• Analysis of software defects to find their root cause and code changes to get correct software behaviour in radio access software for UMTS, LTE • Involved in taking ownership of software components from vendors and start improvements/developments in house – starting from code repository structuring to feature addition, addressing defects etc. • Setting up, configuring and maintaining end-to-end (Mobile handsets to Core network) test setups
Tasks	End-to-end test setup assembling • The test bench spanned from Mobile phones to Core network setting up • Installation, configuration, interfacing of components • Interactions with individual component teams for installations and configuration of their modules and addressing issues experienced during the process • Used OpenStack cloud platform, LxC containers etc. WCDMA Phy source code ownership transition: • Took over the ownership of Phy code from vendor • Interaction with vendor and partner software technical teams • Maintenance of L1 source code from vendor, bug fixing, analysis of performance issues LTE L2 protocol stack: • Analysis of software bugs in L2 protocol stacks • Contributed in implementation and bug fixing in Rel-9 ICIC feature
Team size	One to six depending on activity

**3) Saankhya Labs Pvt. Ltd. (<http://www.saankhyalabs.com>), Pune; (5m)
Senior Member Technical Staff (August 2013 – December 2013)**

	• Optimized hand assembly programming for Saankhya Lab's VLIW processor. • Interface and controller design for integrating third party DL and UL UE layer-1 C code compliant to LTE Rel.-9 specs.
Tasks	3x Sampling Rate Upsampler: • Fixed point C model development for rate 3x upsampler • Optimized assembly language implementation on Saankhya Lab's VLIW DSP Interface and controller design for integrating third party DL and UL

	UE Layer-1 C code compliant to LTE Rel.-9 specs: • Design of messages and their sequences and state machine for LTE Rel-9 downlink and uplink processing • Implementation of state machine frame work for DL in C.
Team size	One to three.

3) Aricent Group(www.aricent.com), Gurgaon; (3y, 3m)

Technical Leader (January 2012 – July 2013)

Senior Software Engineer (April 2010-December 2011)

	• LTE eNB Layer 1 software development for 3GPP Rel-9 TS 36.211, 36.212 • Maintenance of GSM/GPRS/EGPRS BTS DSP software component for a tier 1 Original Equipment Manufacturer.
Tasks	LTE eNB L1 SW Development: • Low level design of modules described in 36.211, 36.212 for eNB DL and UL • Coding of modules in fixed point C • Integration of L1 software components like Femto Forum L1 API for interface with MAC layer (not integration with L2 SW) • Module testing, system component testing, and integrated L1 software testing on Spectrum Digital EVM for 6488, writing test harness in C and test signal generation for testing eNB UL software in Matlab • Porting the L1 Software to TI DSP 6488 and optimization of code to ensure UL and DL executions meet LTE spec imposed deadlines • Writing profiler using time stamps provided by 6488 Chip Support Library • Creation/modification of DSP/BIOS tasks • Writing software modules for interfacing with EDMA3, TCP2 peripherals GSM BTS DSP SW Component maintenance: • Implementation of antenna switching algorithm and guiding two junior colleagues in implementing downlink frequency adjustment algorithm for catering to mobile stations moving at very high speeds for TCI 6488 DSP • DSP Software component testing in an Layer-3 setup • Replication of reported issues in local BTS laboratory • Debugging issues and log analysis for BTS internal software component interfaces and external Abis interface (with BSC) as per GSM call flows • Code base lining with other BTS projects (merging changes on other branches in Clearcase) • Monitoring the network KPIs' variations in accordance with changes in code or reasoning the changes observed with respect to benchmark build
Team size	Two to four.

4) Ikanos Communications (India) Pvt. Ltd. (www.ikanos.com), Bangalore; (2y, 9m)

Member Technical Staff (July 2007- March 2010)

Algorithm development/simulation/coding:	
Time domain equalization for ADSL modems CO and CPE.	
Tasks	- Implemented MMSE TEQ in Matlab using floating point arithmetic - Implemented Levinson Durbin algorithm for Toeplitz matrix inversion in Matlab using floating point arithmetic - Matlab model of Ikanos CO receive path including digital filters, decimators, RFI cancellation notch filters. - Continuous software bug fixes such as ensuring transmit PSD conformance with the standard, no-connects, performance issues such as data rate requirements, noise margin failures etc.
Software testing/ test automation:	
Automation for configuration of CO, CPE , channel simulator and noise generator.	
Tasks	Perl scripts were written and modified to suite the testing requirements. CPAN Modules used included Win32-SerialPort, Net-telnet, Time-HiRes, WriteExcel, ParseExcel etc.

5) Seasolve Inc. (www.seasolve.com), Bangalore; A division of Cranes Software International Limited (www.cranessoftware.com), Bangalore; (1y, 2m)
Software Engineer (May 2006- June 2007)

Implementation of OFDMA PHY of 802.16e-2005	
This is a test and measurement solution for mobile WiMAX.	
Responsibility	- Involvement in understanding of OFDMA PHY of DL PUSC zone of 802.16e-2005 std. - Development of code for Base Station signal transmitter and Mobile Station receiver - Literature survey for PHY features used in mobile WiMAX.
System description	- Involved four modules: (a) Base station transmitter, (b) Impairment simulation, (c) Mobile/subscriber station receiver and (d) Measurements and graph display. - BS transmitter included PHY chain including randomizer, FEC, interleaver, repetition encoder, constellation mapper, distributed sub-carrier permutation (PUSC), OFDMA frame formation including pilot tones, DFT based OFDM modulator were implemented. - Impairments included ETSI UMTS pedestrian-A and vehicular-B mobile channel model, phase noise, IQ imbalance, quadrature skew, DC offset, frequency offset and AWGN - SS receiver included estimation and correction of frequency offset and channel - Measurements module included display of EVM, CCDF, instantaneous power, power spectrum with spectral mask, FCH bit fields, estimated channel frequency response.
Tools	Matlab, LabVIEW
Team size	Three.

6) Wipro Technologies (www.wipro.com), Pune ; (6m)
Project Engineer (Dec. 2005- May 2006)

Involved in self improvement

Studied signal processing algorithms for wireless communications	
Responsibility	- Studied OFDM mode of 802.16-2004 fixed WiMAX - Studied MIMO techniques for wireless such as spatial multiplexing (BLAST), spatial diversity (space time coding), beam-forming, diversity combining techniques at receiver. - Studied E-UTRA and LTE documents and spatial channel models published by 3gpp - Familiarized with DSPs with documentation of TI 64xx series or processors and the IDE Code Composer Studio and C programming
Tools	Turbo C, Code Composer Studio
Team size	One

**6) International Institute of Information Technology, Hyderabad; (Part time, 2y)
Project Assistant (Dec. 2003 – Nov. 2005)**

WLAN Deployment in IIIT-H campus and Hostels	
Responsibility	I was a part of eight member team which deployed 802.11g wireless network in IIIT-Hyderabad campus which consisted of over 250 access points (AP). I was involved in survey of coverage area of the Access Points, empirical data collection for placement of APs, in obtaining the rates achieved at different distances from the AP, inventory record maintenance, providing support for users in configuration of the WLAN cards post deployment and demonstrations of streaming video over the deployed network.

**7) Spectronic Instruments Pvt. Ltd, Mumbai. (6m)
Trainee Service Engineer (April 2002- Sept. 2002)**

Maintenance of Thermal Analysis equipments	
Installation, maintenance measurements instruments.	
Responsibility	- Installation/maintenance of measuring instruments such as differential scanning calorimeter, thermo gravimetric analysis, and particle analyzers etc. - Training the users on new instruments/features in the instruments

MASTER OF SCIENCE DISSERTATION at IIIT-Hyderabad (2005)

Optimal Trade-off Between Covariance Matrix Size and Number of Data Vectors used in Estimation of Covariance Matrix in Blind Channel Identification.

Eigen-subspace method for blind channel identification is based on second order statistics (SOS) of vectorized over sampled received signal and least squares solution for solving the system of linear equations obtained using orthogonality between noise subspace and channel matrix column space. Practically, we have finite number of samples to estimate the covariance matrix of vector received signals. Selection of vector size gives rise to two opposing forces. The larger vector size firstly gives poorer covariance estimate due to lower averaging, which is especially true with vector processors and secondly, larger noise subspace dimension gives better least squares solution for final estimate. In the thesis it is pointed out that there exists an optimum vector size for which the channel estimate variance is minimum using the published results on variance of eigen vector estimates. Extensive Matlab simulations were performed. First order matrix perturbation theory was tried to put the simulation results on analytical foundation. The literature

survey included understanding more than 200 research papers in signal processing, communications and mathematics.

EDUCATIONAL QUALIFICATIONS

- 2003-2005 **International Institute of Information Technology (IIIT), Hyderabad**
M.S., Communication Systems & Signal processing CGPA: 7.8/10
Courses: (i) Digital Communications, (ii) Algebraic Coding Theory,
(iii) Space-Time Wireless Communications, (iv) Adaptive Signal Processing
(v) Pattern Recognition.
- 1997-2001 **Veermata Jijabai Technological Institute (VJTI), Mumbai**
B.E., Electronics Engineering Percentage: 56.43%
Courses: (i) Signals & Systems, (ii) DSP, (iii) Digital Communication, (iv)
Intel 80x86 microprocessors